

RESEARCH ARTICLE



Native woody vegetation in central Argentina: Classification of Chaco and Espinal forests

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Funding information

Raíces Program, Argentinean Ministry of Science and Technology MINCyT

Co-ordinating Editor: José Paruelo

Abstract

Question: What are the composition and spatial patterns of native woody plant communities in the southern Great Chaco and Espinal?

Location: Córdoba Province, central Argentina, an area of ca. 161,000 km².

Methods: We collected 351 geo-referenced relevés representative of the geographic, topographic and ecological variation of the Chaco and Espinal woody vegetation in central Argentina. The relevés were classified into vegetation types using the hierarchical ISOPAM method. Forest and shrubland types were described on the basis of diagnostic species occurrences and their distribution in relation to environmental factors. A map of the actual vegetation derived from remote-sensed images (Landsat) and field data was used to describe the current distribution and abundance of the different vegetation types.

Results: The classification of the 351 plots × 837 species matrix revealed two major clusters comprising seven woody vegetation types corresponding to Chaco lowland and mountain forests and shrublands, Espinal forests and edaphic vegetation. The most important gradients in woody vegetation types are related to elevation, temperature and rainfall variables.

Conclusions: Subtropical seasonally dry woody plant communities from the southern extreme of the Great Chaco and Espinal forests were described for the first time based on complete floristic data. Our results show that lowland Chaco native forests, as well as replacement communities, are still present in its southern distribution range and are well distinguishable from other vegetation types such as the Espinal and mountain forests. Overall, extensive Espinal forests have almost disappeared while Chaco vegetation is highly fragmented and degraded.

KEYWORDS

altitudinal gradient, central Argentina, Espinal, Great Chaco, subtropical xerophytic forest, vegetation maps

1 | INTRODUCTION

The woody vegetation of central Argentina encompasses two major seasonally dry forest ecosystems: the Chaco and Espinal (Burkart, Bárbaro, Sánchez, & Gómez, 1999; Cabrera, 1976). This region,

including the confluence of two main biogeographic units, hosts high levels of biodiversity (Cabido, Funes, Pucheta, Vendramini, & Díaz, 1998; Kuemmerle et al., 2017; Luti et al., 1979). The former belongs to the Great Chaco, an area of ca. 1,200,000 km² in Argentina, Paraguay and Bolivia, comprising xerophytic forests

and shrublands distributed through both lowlands and mountains (Bucher, 1982; Prado, 1993; Zak, Cabido, Cáceres, & Díaz, 2008). The Espinal occupies a narrow belt of about 307,000 km² surrounding the Pampa grasslands and “Campos” in central eastern Argentina (Lewis & Collantes, 1973), comprising mostly forests and shrublands dominated by microphyllous deciduous and thorny species of *Prosopis* (Luti et al., 1979). These regions, the largest seasonally dry forests in South America (Baldi, Houspanossian, Murray, Rueda, & Jobbágy, 2015; Fehlenberg et al., 2017), are among the few areas in the world where the transition between the tropics and the temperate zones does not occur in the form of a desert but as semi-arid forests and shrublands (Morello & Adámoli, 1974; Prado, 1993). Both vegetation units are characterized by a specific vegetation and fauna that not only determine consistent biodiversity values (Cabido et al., 1998; Grau et al., 2015; Nori et al., 2016; Periago & Leynaud, 2009; Prado, 1993; Torrella, Ginzburg, Adámoli, & Galetto, 2013), but also provide numerous ecosystem services (Conti & Díaz, 2013) necessary for the subsistence of local communities and the regional economy (Cáceres & Rodríguez-Bilella, 2014; Zak et al., 2008).

Before the Spanish occupation, the Espinal, and especially the Chaco, regions were covered by primary forests and woodlands, alternating with patches of grasslands maintained through traditional management by Amerindians, who used fire in their hunting practices (Bucher, 1982). This balance between woody and herbaceous vegetation was disrupted when Europeans occupied the area and selectively cleared the forest for extensive livestock raising, a practice that has continued for more than four centuries (Díaz, Acosta, & Cabido, 1994; Luti et al., 1979; Morello & Saravia Toledo, 1959; Schofield & Bucher, 1986). During the 20th century, and especially in more recent decades, the lowland landscape occupied by Chaco and Espinal forests has been transformed into an agricultural system at high deforestation rates (Boletta, Ravelo, Planchuelo, & Grilli, 2006; Britos & Barchuk, 2008; Gasparri & Grau, 2009; Johnson & Zuleta, 2013; Schofield & Bucher, 1986). On the other hand, in the mountain area, fire, grazing and the recent advance of woody exotic species transformed the landscape into a complex mosaic of degraded native forests, monospecific stands of woody exotics, shrublands and grasslands (Argañaraz, Gavier Pizarro, Zak, & Bellis, 2015; Gavier & Bucher, 2004; Giorgis, Cingolani, & Cabido, 2013; Giorgis & Tecco, 2014; Giorgis et al., 2011, 2017; Zeballos, Tecco, Cabido, & Gurvich, 2014). Both areas, include the most important remnants of subtropical xerophytic forests worldwide (Bastin et al., 2017; Hansen et al., 2013; Vallejos et al., 2014), although the data on species composition, despite its outstanding biodiversity, are scarce (Bastin et al., 2017; Kuemmerle et al., 2017). Although some wide phytogeographical studies have been carried out (Kurtz, 1904; Luti et al., 1979), hitherto published surveys describe only partially the species composition of forests and shrublands and encompass very small areas (Cabido, Carranza, Acosta, & Páez, 1991; Cabido, Manzur, Carranza, & González Albarracín, 1994; Lewis, Noetinger, Prado, & Barberis, 2009; Noy-Meir et al., 2012; Sayago, 1969; Zak &

Cabido, 2002). Therefore, synthetic comparative studies of vegetation types based on complete floristic composition across these biogeographic units are needed.

In this article we provide succinct, but comprehensive, analysis of the floristic composition and distribution of the major woody vegetation types in central Argentina (Córdoba region). Specifically, we aim to answer the following questions: (1) what is the floristic composition of the major native woody vegetation types in the southern Great Chaco and Espinal; (2) how do these vegetation types relate to major environmental factors; and (3) what is the current distribution of these native vegetation types?

2 | METHODS

2.1 | Study area

The study area comprises the province of Córdoba (Argentina), which covers ca. 161,000 km² (Figure 1) and encompasses large areas of the two major biogeographic units of seasonally dry forests: the Chaco and Espinal (Burkart et al., 1999; Cabrera, 1956). The area, with an altitudinal gradient spanning from 100 to almost 2,900 m a.s.l., includes two main geomorphologic units: an old mountain system comprising three main ranges, and lowlands filled with Pleistocene and Holocene sediments, distributed both to the east and west of the mountains (Carignano, Kröhling, Degiovanni, & Cioccale, 2014). The mountains run for more than 400 km from north to south and divide the lowlands into two well-differentiated areas: a western semi-arid basin and an eastern sub-humid plain. The northeast and northwest extremes of the area are occupied by saline depressions supporting azonal vegetation. A south-to-north gradient of increasing temperature (15–18°C) and an east-to-west gradient of decreasing annual precipitation (>900–450 mm) occur in the lowlands. There is also a consistent climatic variation following the elevation gradient of the mountains (Tecco et al., 2016); the annual temperature varies from 14°C (at about 500 m a.s.l.) to <7°C (above 2,500 m a.s.l.), while annual precipitation varies from <700 mm to >900 mm. Following the temperature, precipitation and topographic gradients, the climate of the study area ranges from subtropical to warm temperate in a north-to-south direction in the lowlands and from warm temperate to cold temperate in the mountains. Rainfall occurs mainly during the warmer months, with about 80% concentrated between October and March (Capitanelli, 1979).

The lowlands and plains to the north and western parts of the study area belong to the Chaco Phytogeographical Province according to the classification of Cabrera (1976). This sector of the Chaco is part of the Western (semi-arid) Chaco District and corresponds to the southernmost extreme of the Great South American Chaco dominated by *Aspidosperma quebracho-blanco* and *Schinopsis lorentzii*. The central and southern part of the study area belongs to the Espinal phytogeographic Province, more precisely to the Algarrobo (*Prosopis nigra* and *P. alba*) and Caldén (*P. caldenia*) Districts (Cabrera, 1976; Luti et al., 1979). In

the mountains (highest peak 2,979 m a.s.l.), the altitudinal gradient results in the formation of distinct vegetation zones or belts and is part of the Mountain Chaco District (Cabrera, 1976; Luti et al., 1979). However, the upper belt has been related to the Andean and Patagonian biota (Cabido et al., 1998; Martínez, Arana, Oggero, & Natale, 2017).

2.2 | Vegetation sampling and climatic data

A total of 351 geo-referenced floristic relevés dealing with native woody vegetation were sampled to represent the geographic, topographic and ecological heterogeneity found in the Chaco and Espinal forests and shrublands. In our study, only native forests and shrubland plots collected during the last 20 years were analysed, while plots from forestry plantations (mainly *Pinus* species) or plots conspicuously dominated by exotic trees (mainly *Ligustrum lucidum* and *Gleditsia triacanthos*) were not included. Most of the relevés are part of the sPlot as SA-AR-001 database (www.idiv.de/splot). The relevés were performed during the rainy season (summer of Southern Hemisphere); for each 20 m × 20 m plot, the vascular plant species composition was recorded, together with cover-abundance estimates according to Braun-Blanquet scale (Braun-Blanquet, 1932). Cover and height by strata were also estimated; strata included tree (>5 m to canopy height), shrub (0.5–5.0 m) and herb (<0.5 m), although the strata height ranges could be adjusted in the field to reflect local vegetation structure. Exotic and endemic taxa at the national level were identified following Zuloaga, Morrone, and Belgrano (2008) and its on-line updates (www.darwin.edu.ar).

For each plot elevation, precipitation (seasonality and annual precipitation) and temperature (annual mean and range) were estimated by interpolation of data from the WorldClim database (www.worldclim.org;

Hijmans, Cameron, Parra, Jones, & Jarvis, 2005), at a spatial resolution of 2.5 arc-min for continental South America. All mapping and environmental data extraction was conducted using QGIS 2.6.1 (QGIS Development Team 2012). ANOVA models were performed to analyse if environmental variables differed among the vegetation types. Residuals were examined, and found to support the assumptions of normality and homoscedasticity. This analysis was performed in R v 3.2.4 (R Foundation for Statistical Computing, Vienna, Austria).

2.3 | Vegetation analyses and classification of woody plant communities

The ISOPAM method was employed for classification of the data set of 351 plots and 837 species into the major vegetation types and their corresponding diagnostic species groups (Černý et al., 2015; Schmidlein, Tichý, Feilhauer, & Faude, 2010). Hierarchical ISOPAM was run on the Bray-Curtis dissimilarity matrix. The maximum number of clusters on each hierarchical level was arbitrarily set to ten, using a standardized G statistics above five as a measure of species overall capacity to define clusters in a given partition (Schmidlein et al., 2010). To define diagnostic species for each vegetation type, the ϕ coefficient of fidelity was used (Chytrý, Tichý, Holt, & Botta-Dukát, 2002); only those species with $\phi > 0.1$ and a statistically significant association ($p < .01$) with a particular vegetation type according to Fisher's exact test were considered as diagnostic. These analyses were performed in the JUICE 7.0 program (Tichý, 2002).

The vegetation matrix was arranged through isometric feature mapping (ISOMAP) (Černý et al., 2015; Tenenbaum, de Silva, & Langford, 2000), using the number of neighbours to the optimal value from the first hierarchical level of the ISOPAM classification. The environmental factors were related to ISOMAP ordination

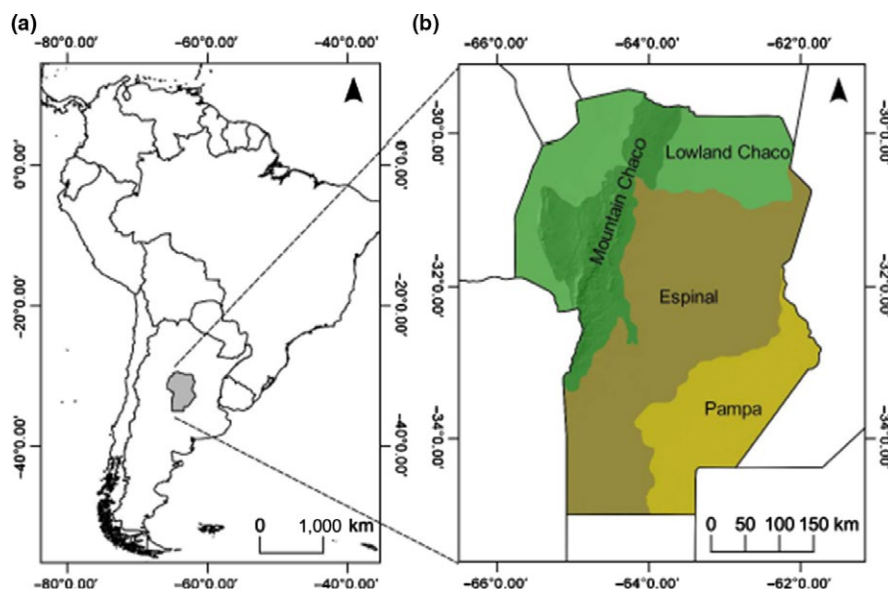


FIGURE 1 (a) Location of Córdoba Province in central Argentina. (b) Phytogeographical units in Córdoba region according to Cabrera (1976)

through the *envift* function from the *vegan* R package. Each vegetation type was named after the species with: (1) a $\phi > 0.2$ and a statistically significant association ($p < .01$), and (2) a constancy of more than 70% and the higher mean cover.

2.4 | Mapping woody vegetation types

Cover of native woody vegetation types was analysed using a vegetation map derived from Landsat TM satellitary imagery acquired in the growing season of 2009 and field data. The vegetation map was obtained through the classification of 12 Landsat 5 TM scenes corresponding to the beginning of the Southern Hemisphere's growing season. Briefly, the vegetation map was produced as follows: first the images were classified using unsupervised classification algorithms; then vegetation field data distributed on areas referable to the vegetation classification obtained through ISOPAM were analysed; finally a supervised classification of the images was performed through a maximum likelihood routine using as training sites a group of areas for which vegetation types had been ascertained in the previous step. The validation of the maps was performed using both field data and visual interpretation of historic orthophotos and "Quickbird" images (Google Earth 2008). The digital map had an overall accuracy of about 90% (Kappa statistic was > 0.88). (For further information on the construction of the map and accuracy assessment see Appendix S2.)

3 | RESULTS

We recorded a total of 837 vascular plant taxa (768 native and 70 exotic species) distributed among 106 families and 473 genera. The dominant families Asteraceae and Poaceae accounted for 32% of all species with 74 and 54 genera, respectively. Other common families were Fabaceae (31 genera), Euphorbiaceae (12), Malvaceae (18), Solanaceae (11) and Cactaceae (11). A total of 113 endemic species were recorded at the national level.

3.1 | Native woody vegetation types and environmental gradients

The results of the ISOPAM classification are summarized in a synoptic table (Table 1, Appendix S1, see also Figure 2). The resulting partitions revealed two main clusters that were further divided into seven woody vegetation types. The first division may be ascribed to phytogeographic units occurring in the study area: four vegetation types included in Cluster 1 (C1) belong to the Mountain Chaco and Espinal vegetation, while the remaining three comprised in Cluster 2 (C2) describe the lowland Chaco forests and scrubs. The average number of species per plot, as well as the proportions of endemics, varied among vegetation types (Table 1). The highest richness was recorded in Type 1.2 *Acacia caven* – *Baccharis aliena* – *Bouteloua curtipendula* mountain Chaco shrubland; the lowest number was found in Type 2.2 *Aspidosperma quebracho-blanco* – *Acacia praecox* –

Ziziphus mistol subtropical Chaco forest. The study area hosted 113 (13.5%) endemic taxa, with the proportions ranging from 20.5% in Type 1.4 *Polylepis australis* – *Deyeuxia hieronymi* sub-Andean forest and shrubland to 7.4% in Type 1.3 *Prosopis nigra* – *Celtis ehrenbergiana* – *Prosopis caldenia* Espinal forests confined to the centre, east and south of the area. The latter type also exhibited the highest proportion of exotic species (14.3%), while Type 2.2 *Aspidosperma quebracho-blanco* – *Acacia praecox* – *Ziziphus mistol* subtropical Chaco forest supported only 3.8% of exotics (see Table 1, Appendix S1).

Differences in floristic composition were related to elevation ($r^2 = .48$, $p = .001$), annual mean temperature (AMT; $r^2 = .58$, $p = .001$), temperature annual range (TAR; $r^2 = .12$, $p = .001$), annual precipitation (AP; $r^2 = .18$, $p = .001$) and precipitation seasonality (PS; $r^2 = .34$, $p = .001$; Figure 2, Table 2). The most influential factor related to changes in composition was annual mean temperature with the lowland Chaco formations (i.e. Types 2.1, 2.2 and 2.3) characterized by high annual mean temperatures (Figure 2). Within mountain vegetation, *Polylepis australis* – *Deyeuxia hieronymi* sub-Andean forest stands (Type 1.4) occupy the cooler sites (Table 2). The elevation gradient mirrored the trends in temperature, discriminating the mountain types (Types 1.1 and 1.2) from the Espinal forest (Type 1.3) and lowland Chaco (Types 2.1, 2.2 and 2.3) vegetation types (Table 2); temperature and elevation also discriminated forest and shrubland mountain Chaco vegetation from *Polylepis australis* – *Deyeuxia hieronymi* sub-Andean forest (Type 1.4). Annual precipitation did not show a wide range overall but significantly differed within the lowland Chaco vegetation types with the *Aspidosperma quebracho-blanco* – *Acacia praecox* – *Ziziphus mistol* subtropical Chaco forest exhibiting the highest annual rainfall and the *Larrea divaricata* – *Acacia gilliesii* – *Mimozyanthus carinatus* subtropical xerophytic Chaco forest and shrubland occupying the drier areas to the west of the study area (Table 2). Annual rainfall did not differ among vegetation types included in Cluster 1. Precipitation seasonality showed only narrow variations among the different vegetation types but discriminated between woody types in the lowland Chaco.

Currently only 21.5% (ca. 36,094 km²) of the province is covered by woody vegetation (Figure 3). Espinal vegetation is distributed in highly fragmented relict patches covering ca. 4,934 km². Even though Chaco vegetation is fragmented and degraded, it is still present in the territory comprising 31,160 km², mainly distributed in the western plains and in the mountains.

3.2 | Cluster 1

3.2.1 | Type 1.1 *Lithraea molleoides* – *Croton lachnostachyus* mountain Chaco forest

This vegetation type, occupying 2,900 km², was sampled in a relatively wide altitudinal range in the mountains, from 600 to 1,400 m a.s.l. (Table 2). The physiognomy of this type is that of a low, open to locally closed forest, with a total cover between 80 and 100%. Tree taxa such as *Lithraea molleoides*, *Condalia buxifolia*, *Myrcianthes cisplatensis* and *Zanthoxylum coco*, together with some

TABLE 1 Synoptic table showing the identified woody vegetation types along with the percentage constancy and mean cover (superscript) based on 351 relevés collected in Córdoba, central Argentina. Species are sorted by decreasing fidelity within types. Dark grey and light grey indicate $\phi > 0.2$ and $\phi > 0.1$, respectively. Vegetation types are: 1.1: *Lithraea molleoides* – *Croton lachnostachyus* mountain Chaco forest; 1.2: *Acacia caven* mountain Chaco shrubland; 1.3: *Prosopis nigra* – *Celtis ehrenbergiana* – *Prosopis caldenia* Espinal forest; 1.4: *Polylepis australis* – *Deyeuxia hieronymi* sub-Andean forest and shrubland; 2.1: *Larrea divaricata* – *Acacia gilliesii* – *Mimozyanthus carinatus* subtropical xerophytic Chaco forest and shrubland; 2.2: *Aspidosperma quebracho-blanco* – *Acacia praecox* – *Ziziphus mistol* subtropical Chaco forest; and 2.3: *Atriplex argentina* – *Cyclolepis genistoides* – *Maytenus vitis-idaea* halophytic/sub-halophytic shrubland. Life forms (LF): c, cactus; cl, climber; e, epiphyte; g, grass; gr, graminoid; h, herb; k, succulent; p, palm; s, shrub; ss, subshrub; t, tree. Symbols: †, endemic species; ‡, exotic species.

Cluster		C1				C2		
Vegetation type		1.1	1.2	1.3	1.4	2.1	2.2	2.3
Number of relevés		48	60	64	27	106	17	29
Average species number		49.3	70.7	40.1	38.9	33	23.5	34.1
		± 1.4	± 2.3	± 1.8	± 1.7	± 0.9	± 2.8	± 2.4
N° of endemic species		25	60	20	32	42	12	23
N° of exotic species		28	54	34	6	15	4	14
N° of native species		291	410	237	150	204	105	194
Diagnostic species		LF						
<i>Lithraea molleoides</i>	t	79 ³	48 ²	2 ¹	.	.	.	.
<i>Myrcianthes cisplatensis</i>	t	25 ³	5 ²	.	.	.	.	.
<i>Condalia buxifolia</i>	t	35 ³	8 ²	3 ¹	.	.	.	.
<i>Croton lachnostachyus</i>	s	92 ²	32 ¹	14 ¹	.	5 ¹	41 ¹	.
<i>Oplismenus hirtellus</i>	g	44 ²	.	3 ³	.	.	.	.
<i>Lorentzianthus viscidus</i>	s	67 ²	10 ¹	6 ¹	.	3 ¹	.	.
<i>Ruprechtia apetala</i>	t	56 ²	33 ¹	3 ¹	.	22 ²	12 ²	.
<i>Condalia montana</i> †	t	60 ²	30 ¹	9 ¹	.	1 ⁺	6 ⁺	.
<i>Zanthoxylum coco</i>	t	60 ²	25 ²	8 ³	.	.	.	.
<i>Setaria oblongata</i>	g	67 ²	10 ¹	2 ³	.	1 ⁺	.	.
<i>Lantana camara</i>	s	19 ²	5 ²	.	.	.	.	.
<i>Gouinia latifolia</i>	g	40 ¹	7 ¹	9 ¹	.	2 ¹	.	.
<i>Acacia caven</i>	s	58 ¹	92 ²	28 ¹	.	6 ¹	6 ⁺	3 ⁺
<i>Baccharis aliena</i>	s	4 ¹	35 ²	.	44 ¹	.	.	.
<i>Acalypha communis</i>	ss	25 ¹	67 ²	3 ¹	.	.	.	.
<i>Flourensia oolepis</i> †	s	4 ¹	13 ³	.	.	.	.	.
<i>Acanthostyles buniifolius</i>	s	2 ¹	50 ¹	.	4 ⁺	.	.	.
<i>Bouteloua curtipendula</i>	g	27 ¹	73 ¹	6 ¹	4 ⁺	9 ²	.	.
<i>Jarava pseudoichu</i>	g	40 ¹	57 ²	30 ³	11 ¹	1 ¹	12 ⁺	3 ¹
<i>Colletia spinosissima</i>	s	8 ¹	22 ²	.	37 ⁺	.	.	.
<i>Nassella filiculmis</i>	g	2 ⁺	13 ²	.	.	.	.	.
<i>Piptochaetium montevidensis</i>	g	4 ⁺	50 ¹	8 ¹	4 ⁺	.	.	.
<i>Nassella cordobensis</i> †	g	8 ¹	38 ¹	2 ⁺	.	1 ⁺	.	.
<i>Baccharis linearifolia</i>	s	2 ¹	35 ¹	.	.	.	.	.
<i>Eustachys retusa</i>	g	4 ¹	57 ¹	6 ¹	.	.	.	.
<i>Vernonanthura nudiflora</i>	ss	6 ¹	43 ¹	.	.	.	.	.
<i>Bouteloua megapotamica</i>	g	.	20 ¹	.	.	1 ⁺	.	.
<i>Aristida spagazzinii</i>	g	.	18 ¹	.	.	.	.	.
<i>Prosopis caldenia</i> †	t	.	.	23 ³	.	.	.	.
<i>Celtis ehrenbergiana</i>	t	85 ²	65 ¹	89 ²	.	72 ¹	100 ²	24 ⁺

(Continues)



TABLE 1 (Continued)

Cluster		C1				C2		
Vegetation type		1.1	1.2	1.3	1.4	2.1	2.2	2.3
<i>Prosopis nigra</i>	t	.	8 ²	45 ²	.	.	82 ⁺	10 ⁺
<i>Schinus fasciculatus</i>	s	33 ¹	60 ¹	61 ²	.	7 ⁺	47 ¹	28 ¹
<i>Araujia odorata</i>	cl	10 ⁺	17 ⁺	64 ¹	.	7 ⁺	24 ⁺	.
<i>Morus alba</i> †	t	19 ¹	3 ⁺	25 ²	.	.	.	.
<i>Clematis montevidense</i>	cl	38 ¹	40 ¹	92 ¹	.	3 ⁺	47 ⁺	14 ⁺
<i>Nassella hyalina</i>	g	4 ⁺	7 ¹	19 ²	.	1 ³	.	.
<i>Chenopodium album</i> †	h	8 ²	7 ⁺	23 ²	.	1 ⁺	.	.
<i>Leonurus japonicus</i> †	h	23 ¹	7 ¹	31 ²	.	.	.	.
<i>Bromus catharticus</i>	g	10 ¹	12 ¹	36 ¹	30 ⁺	2 ³	.	.
<i>Lycium chilense</i>	s	.	.	25 ¹	.	.	.	.
<i>Polylepis australis</i> †	t	.	.	.	100 ⁴	.	.	.
<i>Blechnum penna-marina</i>	h	2 ⁺	.	.	56 ³	.	.	.
<i>Deyeuxia hieronymi</i>	g	.	.	.	89 ²	.	.	.
<i>Clinopodium odorum</i>	s	.	.	.	67 ²	.	.	.
<i>Maytenus boaria</i>	t	.	.	.	82 ¹	.	.	.
<i>Gaultheria poeppigii</i>	s	.	.	.	82 ¹	.	.	.
<i>Sorghastrum pellitum</i>	g	.	3 ¹	.	37 ²	.	.	.
<i>Stevia achalensis</i> †	ss	.	.	.	100 ⁺	.	.	.
<i>Polystichum pycnolepis</i>	h	.	.	.	63 ¹	.	.	.
<i>Uncinia phleoides</i>	gr	.	.	.	33 ¹	.	.	.
<i>Poa stuckertii</i> †	g	.	.	.	37 ¹	.	.	.
<i>Berberis hieronymi</i> †	s	.	.	.	100 ¹	.	.	.
<i>Larrea divaricata</i>	s	.	5 ¹	6 ¹	.	88 ²	29 ¹	3 ⁺
<i>Acacia gilliesii</i>	s	4 ¹	5 ¹	.	.	74 ²	6 ⁺	.
<i>Justicia squarrosa</i>	ss	8 ¹	.	2 ¹	.	54 ²	18 ²	.
<i>Mimozyanthus carinatus</i>	s	.	.	.	.	76 ²	.	24 ²
<i>Selaginella sellowii</i>	h	4 ²	42 ²	3 ¹	22 ⁺	39 ²	29 ⁺	.
<i>Prosopis flexuosa</i>	t	.	3 ²	.	.	68 ¹	6 ⁺	24 ¹
<i>Cercidium praecox</i>	t	.	.	.	.	68 ¹	41 ⁺	21 ¹
<i>Neobouteloua lophostachya</i>	g	.	.	2 ⁺	.	55 ¹	6 ⁺	.
<i>Jarava ichu</i>	g	2 ¹	.	25 ¹	33 ¹	.	47 ²	7 ⁺
<i>Aspidosperma quebracho-blanco</i>	t	19 ¹	13 ¹	16 ²	.	86 ²	94 ²	35 ¹
<i>Acacia praecox</i>	t	33 ²	7 ¹	5 ²	.	22 ²	88 ²	3 ²
<i>Prosopis kuntzei</i>	t	.	.	2 ⁺	.	.	53 ¹	3 ⁺
<i>Schinopsis lorentzii</i>	t	.	.	.	.	.	35 ¹	.
<i>Ziziphus mistol</i>	t	2 ⁺	3 ¹	5 ¹	.	42 ¹	82 ¹	3 ⁺
<i>Atriplex argentina</i> †	s	.	.	.	.	1 ⁺	.	76 ²
<i>Cyclolepis genistoides</i>	s	.	.	.	.	2 ⁺	.	76 ²
<i>Maytenus vitis-idaea</i>	k	2 ¹	.	.	.	8 ¹	.	66 ²
<i>Sarcocornia ambigua</i>	k	.	.	3 ⁺	.	2 ³	.	31 ²
<i>Allenrolfea patagonica</i> †	k	.	.	.	.	.	12 ⁺	38 ²
<i>Plectrocarpa tetraacantha</i> †	s	.	.	.	.	1 ⁺	.	48 ²

(Continues)

TABLE 1 (Continued)

Cluster		C1				C2		
Vegetation type		1.1	1.2	1.3	1.4	2.1	2.2	2.3
<i>Spartina spartinae</i>	g	.	.	.	.	.	.	17 ²
<i>Suaeda divaricata</i>	k	.	.	.	.	1 ²	.	31 ²
<i>Stetsonia coryne</i>	c	2 ⁺	5 ¹	2 ¹	.	18 ¹	47 ¹	35 ²
<i>Trichloris pluriflora</i>	g	.	7 ¹	3 ¹	.	33 ¹	29 ⁺	59 ¹
<i>Prosopis reptans</i>	s	.	.	.	.	.	.	52 ¹
<i>Geoffraea decorticans</i>	t	8 ¹	13 ¹	91 ³	.	39 ¹	71 ¹	93 ²
<i>Lycium tenuispinosum</i>	s	.	.	.	.	5 ¹	.	66 ¹

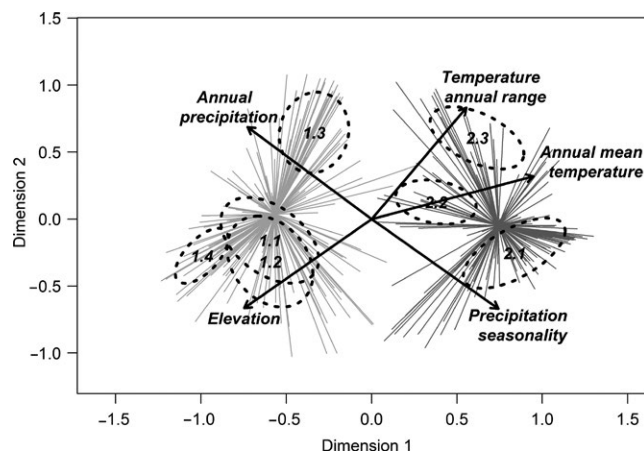


FIGURE 2 Isometric feature mapping plot (ISOMAP), based on Bray-Curtis similarity of 351 plots $\times$ 837 vegetation species matrix for Córdoba province, central Argentina. Vegetation types are those reported in Figure 3. Environmental variables significantly correlated with the sample scores in the ordination space are also reported

shrubs and vines are among the diagnostic species. An open shrub layer is formed by *Croton lachnostachyus* and other species; in the herb layer, the grasses *Oplismenus hirtellus*, *Setaria oblongata* and *Gouinia latifolia* occur with the highest constancy. This vegetation type comprises a total of 291 plant taxa, and species richness is high (49.2 ± 1.4 species per plot). The number of endemic species occurring in this type is 25 (7.8% of the total) and exotic species add up to 28.

3.2.2 | Type 1.2 *Acacia caven* mountain Chaco shrubland

This vegetation type occupies 12,763 km² and is widely distributed across an altitudinal gradient ranging from 400 to 1,700 m a.s.l. in all orientations and on different rock substrates (Figure 3). The general appearance of this vegetation is that of an open and low to medium tall (2–4 m) shrubland with varying cover of the herb layer. This vegetation type is especially rich in herbs and shrubs comprising the bulk of diagnostic species, most of them shared with the mountain forests. The shrubland is alternatively dominated by different species and a variety of shrubs such as *Flourensia olepis*, *Flourensia*

campestris, *Acanthostyles buniifolium*, *Colletia spinosissima* and *Aloysia gratissima*. Individuals of tree species such as *Lithraea molleoides*, *Zanthoxylum coco*, *Schinopsis marginata*, *Ruprechtia apetala* and the palm tree *Trithrinax campestris* are all frequent. Among many grass species, *Jarava pseudoichu*, *Bouteloua curtipendula*, *Eustachys retusa* and *Paspalum notatum* may be dominant in the lowest strata. Species richness was the highest recorded among all the stands included in this study (70.6 ± 2.3 species per plot), while the percentage of endemic and exotic taxa reach 12.9% and 13.1%, respectively.

3.2.3 | Type 1.3 *Prosopis nigra* – *Celtis ehrenbergiana* – *Prosopis caldenia* Espinal forests

This community occurs as isolated patches on the centre, east and southwest plains of the study area (Figure 3) and occupies 4,934 km². Vegetation is represented through a mixture of forests and secondary communities. Trees such as *Prosopis nigra*, *P. caldenia*, *Celtis ehrenbergiana*, *Schinus fasciculatus* and the exotic *Morus alba*, as well as the climbers *Araujia odorata* and *Clematis montevidensis* are among the diagnostic taxa (Table 1). Forest stands to the north and east of the area covered by this type are dominated by *P. nigra*, *Celtis ehrenbergiana*, *Morus alba*, *Geoffraea decorticans* and, only locally, by *P. alba*. In the southern extreme of its range, on sandy soils and old dunes, the canopy is dominated by *P. caldenia*. Species richness is intermediate (40.6 ± 1.8 species plot⁻¹), while the number of endemics is 43. The percentage of exotic species is the highest among the mountain and lowland types (14.3%); some exotic trees (e.g. *Morus alba* and *Ligustrum lucidum*) may dominate in stands surrounded by cultivated land. Conversely, the percentage of endemic species is the lowest of all the vegetation types described (7.4%).

3.2.4 | Type 1.4 *Polylepis australis* – *Deyeuxia hieronymi* sub-Andean forest and shrubland

This 53 km² vegetation type encompasses low forests and shrublands occurring in stands recorded mainly in ravines and deep valleys between 1,600 and 2,400 m a.s.l. The vegetation is dominated by *Polylepis australis*, a small tree widely distributed through the mountains of northwest Argentina, reaching the southernmost distribution of the genus in the study area. Another important

canopy species is *Maytenus boaria*, a small tree forming pure stands in low mountain slopes in the Andes from southern and central Argentina. In addition to the trees, several shrubs (*Berberis hieronymi*, *Gaultheria poeppigii*, *Clinopodium odorum*), herbs (*Stevia achalensis*), grasses and graminoids (*Deyeuxia hieronymi*, *Poa stuckertii*, *Uncinia phleoides*) and ferns (*Blechnum penna-marina*, *Polystichum pycnolepis*) were diagnostic species (Table 1). Species richness was relatively low (38.8 ± 1.7 species per plot), while this forest type exhibited a high proportion of endemics at the national level (20.5%), most of them also occurring in the high mountains from northwest Argentina. In contrast, only six exotic species have been recorded.

3.3 | Cluster 2

3.3.1 | Type 2.1 *Larrea divaricata* – *Acacia gilliesii* – *Mimozyanthus carinatus* subtropical xerophytic Chaco forest and shrubland

This vegetation type occupies 12,877 km² and encompasses a mixture of tall and closed to open forests, shrublands and scrubs with emergent trees. This type was distributed in the northwestern and western semi-arid plains of the study area (Table 2, Figure 3). The

tree layer is represented only by scattered emergent trees, while the shrub layer is largely dominant. Among the diagnostic species, *Prosopis flexuosa* is the main tree emerging from the canopy, while the shrub layer is alternatively dominated mainly by *Larrea divaricata*, *Acacia gilliesii*, *Mimozyanthus carinatus* and *Cercidium praecox*. Some species of Cactaceae such as *Stetsonia coryne* are frequent. Species richness is low in this vegetation type (33 ± 0.9 species per plot). Forty-two endemic species were recorded and the number of exotics is relatively low (15) if compared with the Mountain Chaco forest and shrubland and Espinal forest types.

3.3.2 | Type 2.2 *Aspidosperma quebracho-blanco* – *Acacia praecox* – *Ziziphus mistol* subtropical Chaco forest

This vegetation type covered 1,296 km² in the lowlands of the northern part of the study area (Figure 3). This is the most characteristic forest community in the arid and semi-arid Chaco. The general appearance of this vegetation is that of an open to closed (30%–60% cover) canopy forest with several shrubs forming a high scrub or low tree stratum. The tree layer is characterized by the dominance of *Aspidosperma quebracho-blanco*, an evergreen xerophyllous tree widely distributed through central and northern Argentina, jointly with other

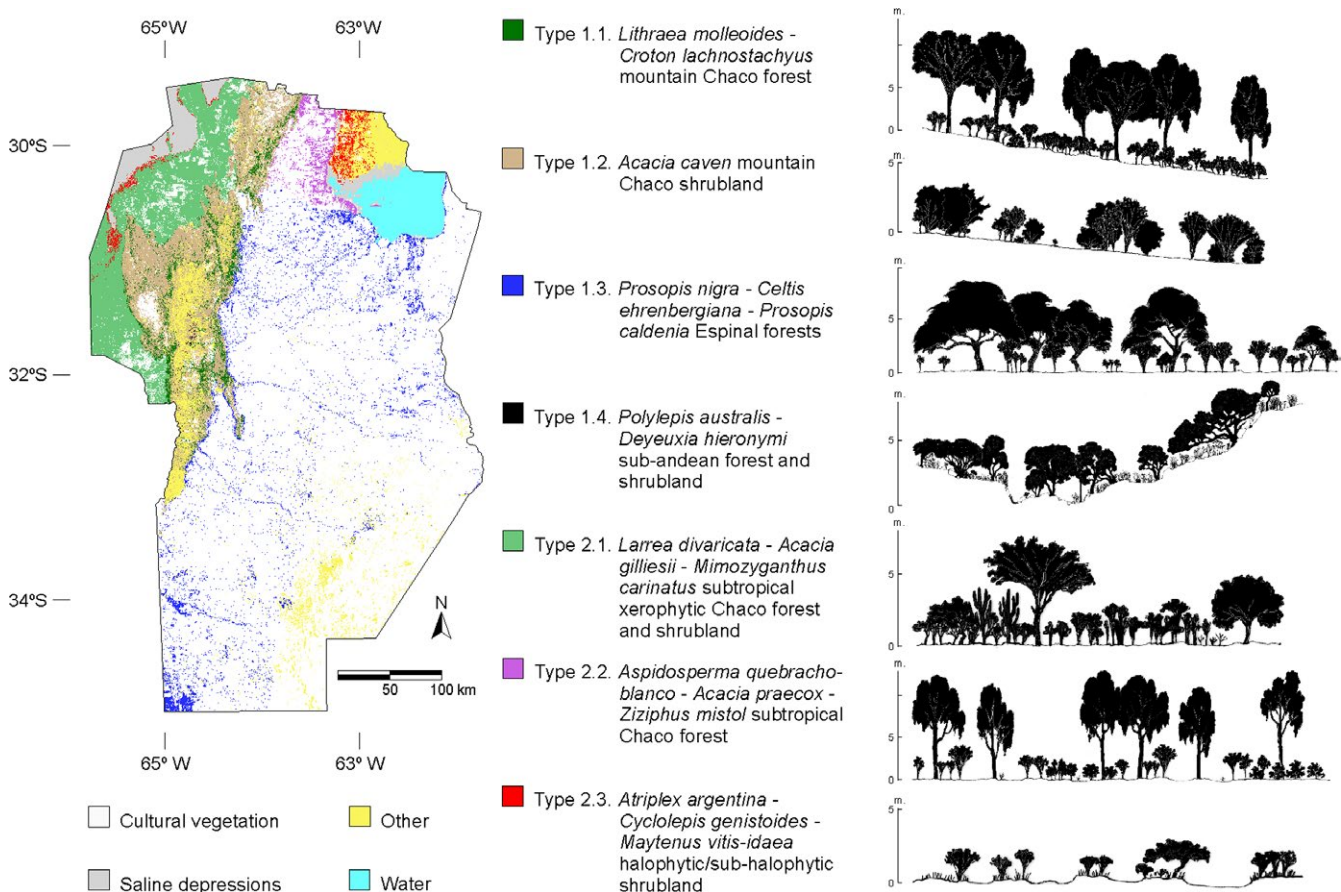


FIGURE 3 Actual vegetation map of Córdoba province derived from remote sensed images (Landsat TM, 2009) and field data. The distributions of the seven vegetation types recognized through ISOPAM classification and their corresponding schematic profiles resembling the physiognomy of the different vegetation types are reported

TABLE 2 Synthetic description of the environmental variables ($\pm$ SE) characterizing the vegetation types present in central Argentina. Vegetation type code names are reported in Table 1

	Elevation (m a.s.l.)	Annual mean temperature ($^{\circ}$ C)	Temperature annual range ($^{\circ}$ C)	Annual precipitation (mm)	Precipitation seasonality (days)
Type 1.1	814 $\pm$ 32.22 ^b	15.66 $\pm$ 0.19 ^d	26.87 $\pm$ 0.1 ^{cd}	682 $\pm$ 8.23 ^{bc}	66 $\pm$ 0.36 ^c
Type 1.2	876 $\pm$ 28.99 ^b	15.58 $\pm$ 0.14 ^d	27.26 $\pm$ 0.09 ^{cd}	664 $\pm$ 7.86 ^{bc}	66 $\pm$ 0.31 ^c
Type 1.3	311 $\pm$ 21.09 ^c	16.92 $\pm$ 0.06 ^c	28.7 $\pm$ 2.8 ^d	705 $\pm$ 13.64 ^b	59 $\pm$ 0.71 ^d
Type 1.4	1959 $\pm$ 29.06 ^a	10.6 $\pm$ 0.13 ^e	26.7 $\pm$ 0.02 ^d	670 $\pm$ 0.38 ^b	65 $\pm$ 0.09 ^c
Type 2.1	377 $\pm$ 14.12 ^c	19.3 $\pm$ 0.12 ^a	28.1 $\pm$ 0.06 ^b	535 $\pm$ 5.56 ^e	73 $\pm$ 0.32 ^a
Type 2.2	260 $\pm$ 26.81 ^{cd}	18.42 $\pm$ 0.18 ^b	27.7 $\pm$ 0.13 ^{bc}	774 $\pm$ 17.46 ^a	66 $\pm$ 0.6 ^c
Type 2.3	149 $\pm$ 13.62 ^d	19.45 $\pm$ 0.13 ^a	28.24 $\pm$ 0.06 ^{ab}	647 $\pm$ 32.73 ^c	70 $\pm$ 1.27 ^b

Different letters indicate significant differences among vegetation types according to post-hoc Tukey contrasts for multiple comparisons.

species such as *Prosopis nigra*, *Acacia praecox* and *Ziziphus mistol*, which may locally dominate after the removal of the tree canopy. Some stands confined to the southeast also comprise other tree species like *Prosopis kuntzei*, *Celtis ehrenbergiana* and the palm *Trithrinax campestris* (see Appendix S1). Some other stands are characterized by the presence of *Schinopsis lorentzii* with emergent individuals taller than 15 m. Towards the transitional areas to saline depressions or to mountain ranges, the columnar cactus *Stetsonia coryne* modifies the physiognomy of this forest. The understorey is alternatively dominated by *Larrea divaricata*, *Capparis atamisquea*, *Condalia microphylla*, *Moya spinosa*, *Castella coccinea* and other shrubs (see Appendix S1). The herb layer is generally sparse. Species number is lower than in any other vegetation types described in the study area (23.53 ± 2.8 species per plot), while the proportion of endemics reaches 11%.

3.3.3 | Type 2.3 *Atriplex argentina* – *Cyclolepis genistoides* – *Maytenus vitis-idaea* halophytic/sub-halophytic shrubland

This vegetation type occupies 1,265 km² and includes all the shrub and scrub communities occurring in the saline depressions located to the northeast and northwest of the study area, in the transitional area between the Chaco forests and the exposed salty “playa” surface, and in scattered halo-hydromorphic stands (Figure 3). Physiognomically, the community in these sites is often composed of a nearly uniform stand of succulent, more or less spinescent, microphyllous to aphyllous shrubs about 1 m tall with open or very open vegetation cover. The diagnostic species of this type are *Allenrolfea patagonica*, *Suaeda divaricata*, *Sarcocornia ambigua*, *Stetsonia coryne*, *Prosopis reptans*, *Geoffroea decorticans*, *Lycium tenuispinosum* and *Spartina spartinae*. *Allenrolfea patagonica* characterizes the vegetation of low elevation, wet, high salinity habitats along the margins of both saline depressions. In more elevated areas, a more dense scrub can be dominated alternatively by the shrubs *Cyclolepis genistoides* and *Maytenus vitis-idaea*. Species richness is low, with an average of 34 ± 2.4 species per plot, while the percentage of endemics is 11.1%. Exotics are few (14), perhaps reflecting the low invasibility of this habitat characterized by harsh environmental conditions.

4 | DISCUSSION

4.1 | Vegetation types and environmental parameters

Our vegetation classification synthesized phytosociological information of native forests and shrublands widely distributed in central Argentina defining two main clusters and seven vegetation types for the Chaco and Espinal biogeographic regions. Woody Chaco and Espinal vegetation types are well separated along a temperature and elevation environmental gradient. Within Cluster 1 (Mountain Chaco and Espinal types), trends in floristic composition were mainly related to elevation, while within Cluster 2 (lowland Chaco vegetation types), composition was driven by shifts in annual rainfall.

Polylepis australis – *Deyeuxia hieronymi* sub-Andean forest and shrubland (Type 1.4) occur at the coldest upper extreme, whereas *Lithraea molleoides* – *Croton lachnostachyus* mountain Chaco forest (Type 1.1) and *Acacia caven* mountain Chaco shrubland (Type 1.2) occupy the lower and warmer altitudinal levels. Thus, our results confirm that the distribution of mountain Chaco vegetation (both forests and shrublands) and *Polylepis* forests and shrublands in central Argentina mountains is strongly driven by temperature shifts following the altitudinal range along more than 1,600 m. In agreement with our findings, Cabido et al. (1998) have reported that, despite a substantial overlap in composition between the different altitudinal levels, Andean elements tend to predominate at higher altitudes in the *Polylepis australis* – *Deyeuxia hieronymi* forest and shrubland, while taxa distributed in southern Brazil and north-eastern Argentina are common at lower altitudes in the mountains. Martínez et al. (2017) have recently proposed a new biogeographic unit for the area where the *Polylepis australis* – *Deyeuxia hieronymi* forest occurs based on the distribution of plants and animals (i.e. Comechingones Province according to the authors), separating it from the Mountain Chaco phytogeographic district where it was formerly included by Cabrera (1976). *Polylepis* forests also occur in the high mountains from northwest Argentina (Cabrera, 1976), where they occupy similar habitats in ravines, deep valleys and steep slopes, and have some species in common with the forests and shrublands described in this study.

With respect to mountain Chaco forests and shrublands (Types 1.1 and 1.2), as shown in Table 2, environmental variables did not differ substantially. Perhaps the differences detected through the floristic analysis may be related to disturbances such as fire and grazing; the occurrence of isolated trees in shrubland stands may support this interpretation. Giorgis et al. (2013, 2017) have already reported a strong effect of fire and grazing on the structure and composition of mountain vegetation, suggesting that floristic and physiognomic discontinuities between mountain forests and shrublands may respond to disturbance and not to environmental factors. While floristic differences between *Polylepis* forest and mountain Chaco vegetation types were clearly related to changes in environmental conditions associated with the elevational gradient, *Lithraea molleoides* forest (Type 1.2) and *Acacia caven* shrubland (Type 1.1) occurred at similar altitudinal and climatic locations and did not seem to constitute different vegetation belts, but rather different sucesional stages induced by disturbances such as fire, grazing and logging. Thus, our classification of the mountain vegetation types is only partially consistent with the traditional scheme of altitudinal vegetation belts, this mainly based on physiognomy, along the mountain Chaco in Argentina (Cabido et al., 1991; Cabrera, 1976; Giorgis et al., 2017; Kurtz, 1904).

Concerning the differences in composition among lowland Chaco vegetation types, our results show a significant relationship with annual precipitation. Stands distributed to the east of the mountain range in the northern part of the study area (Type 2.2 *Aspidosperma quebracho-blanco* – *Acacia praecox* – *Ziziphus mistol* subtropical Chaco forest) showed the highest rainfall records, while the lowest values were recorded towards the west (Type 2.1 *Larrea divaricata* – *Acacia gilliesii* – *Mimozyanthus carinatus* subtropical xerophytic Chaco forest and shrubland). This result is in agreement with the pattern found by Luti et al. (1979) in which a sub-humid (eastern) unit is discriminated from a dry (western) Chaco unit. On the other hand, the distribution of Type 2.3 *Atriplex argentina* – *Cyclolepis genistoides* – *Maytenus vitis-idaea* halophytic/sub-halophytic shrubland, is strongly associated with edaphic factors (Cabido et al., 1994; Ragonese, 1951; Sayago, 1969).

4.2 | Vegetation types and floristic patterns

The ISOPAM classification separated lowland Chaco woody formations (Cluster 2) from mountain Chaco and Espinal vegetation (Cluster 1). This may be a rather surprising result since the Espinal forests have been traditionally defined as an “impoverished Chaco”, with both units sharing many plant species (Cabrera, 1976; Lewis & Collantes, 1973); thus, a higher similarity between Espinal and lowland Chaco forests could have been expected. As discussed above, perhaps the climatic similarities between the Espinal and the low mountain habitats could explain this unexpected pattern. Additionally, the wide geographic contact between the Espinal forests and the low mountain Chaco vegetation could further explain this finding. Overall, our results reflected complex phytogeographic and floristic relations between the woody vegetation types

described. Further studies should involve other areas of the Great Chaco (both lowland and mountain Chaco) and Espinal in order to achieve a more comprehensive scenario.

Our results show that both mountain Chaco vegetation types exhibited the highest average number of species per plot. This pattern may be related to the environmental heterogeneity typical of mountain habitats and also to the transitional character of low mountain communities. The high average number of species that characterizes some of the identified vegetation types is probably related to the relatively long evolutionary history of these ecosystems compared with temperate forests elsewhere. In fact, the lack of severe glaciations in central Argentina (Prado, 1993) and severe recent tectonic processes (Carignano et al., 2014) may have contributed either to the biodiversity enhancement or the reduced number of exclusive species in each vegetation type (although the *Polylepis* forest and the halophytic vegetation are exceptions). Consequently, most of the diagnostic species for a given type were also recorded in the other types.

Of the 113 nationally endemic species observed, less than ten are restricted to the study area (Chiapella & Demaio, 2015). The vegetation type hosting the higher number of narrow endemics (restricted to the study area) is the *Polylepis australis* – *Deyeuxia hieronymi* forest (five species). This vegetation type also exhibits the higher percentage of endemics at the national level (20.5%). This pattern may reflect the isolation of this vegetation type, distributed at higher altitudes, compared to the other types, occurring at lower altitudes or in the plains. Many studies focusing on the detection of areas of endemism (e.g. Crisp, Laffan, Linder, & Monro, 2001; Murray-Smith et al., 2009; Myers, Mittermeier, Mittermeier, Fonseca, & Kent, 2000) have recognized that a substantial number of endemic species in a given geographic region often correlates with age and isolation of the area, since these factors influence both the evolution (the formation and development of new taxa) and survival (the permanence of endemic relicts; Lesica, Yurkewucz, & Crone, 2006). Mountainous regions are especially rich in endemic plant species with restricted distribution, since those areas represent discontinuities in soil conditions and topography that may promote differentiation in plant populations (Cabido et al., 1998; Martínez et al., 2017).

The Espinal and low mountain Chaco vegetation types seem to be the most invaded. Although we did not include patches dominated by exotic species in this study, we observed that in the Espinal the exotics *Morus alba* and *Leonurus japonicus* were diagnostic species. In addition, from the 34 invasive woody exotic species recorded in the study area (Giorgis & Tecco, 2014), 14 were recorded in this study, comprising the Espinal and the Chaco mountain forests and shrublands. Nowadays the vegetation of central Argentina is not only seriously threatened by habitat loss and fragmentation (Carranza et al., 2014), but also by the establishment of exotic species (Giorgis et al., 2011, 2016, 2017; Hoyos et al., 2010; Lewis et al., 2009; Noy-Meir et al., 2012; Tecco et al., 2016; Zeballos et al., 2014). Therefore, increasing human modifications of these environments are probably irreversibly changing these systems into novel ecosystems through severe modifications both in the abiotic conditions and in the biotic composition (Hobbs, Higgs, & Harris, 2009).

4.3 | The current distribution of native woody types and management recommendations

The actual vegetation map depicts a native remnant Chaco and Espinal vegetation in a highly fragmented landscape (Figure 3), covering only 21.5% of the study area with 31,160 km² and 4,934 km², respectively. According to Kurtz (1904), Sayago (1969) and Luti et al. (1979) and the information therein, a significant part of the study area was covered with woody vegetation (mostly forests but also shrublands and scrubs) at the end of 19th century (only 2,343 km² were cultivated in 1888, as reported in the first National Census). As has also been recently reported by Zak et al. (2008) and Hoyos et al. (2013), the original lowland forests have been severely modified through intensive land use. Agricultural areas and cattle grazing zones cover around 70% of the study area (see the vegetation map in Figure 3), and much of the remaining lowland forest is now seriously degraded and fragmented (Carranza et al., 2014; Frate, Acosta, Cabido, Hoyos, & Carranza, 2015). In the mountain area, due to the strong influence of grazing and fire activities, the remaining forest patches are restricted to inaccessible zones such as ravines and deep valleys (Cingolani, Renison, Tecco, Gurvich, & Cabido, 2008).

In the Espinal area (Type 1.3) the conversion of primary forests to agricultural land began at rather high rates at the end of the 19th century and followed a similar trend until approximately the First World War. As a result, only <5,000 km² of mostly degraded forest remain and the invasion of alien trees, also reported by Lewis et al. (2009) and Noy-Meir et al. (2012), is a real threat. The *Aspidosperma* subtropical Chaco forest (Type 2.2) has been recently converted to agriculture lands (Hoyos et al., 2013; Zak et al., 2008), mainly due to the advance of soybean cultivation. Only 1,296 km² of this forest still remains and would be more endangered if international demand and prices of commodities further stimulate the expansion of agriculture in marginal lands. These trends have also been reported for other parts of the Great Chaco in northern Argentina (Boletta et al., 2006; Gasparri & Grau, 2009; Grau, Gasparri, & Aide, 2005), and also in Paraguay (Caldas, Goodin, Sherwood, Campos Krauer, & Wisely, 2015) and Bolivia (van Gils & Ugon, 2006). Non-irrigated agriculture is still not profitable in the drylands occupied by the *Larrea* xerophytic Chaco shrublands (Type 2.1), perhaps explaining the relative high cover of this vegetation type (almost 13,000 km²). Studies on vegetation succession of *Larrea* sp. formations indicate that they may evolve to xerophytic forests and woodlands when the frequency and intensity of disturbance are reduced (Cabido et al., 1994), suggesting a high conservation value of this vegetation type. Mountain Chaco forests (Type 1.1) have also been deforested at high rates (Zak, Cabido, & Hodgson, 2004), with only 2,900 km² remaining. Although the *Acacia caven* mountain Chaco shrubland (Type 1.2) still covers a relatively large area (12,763 km²), together with the *Lithraea* forest, it is severely threatened by the invasion of exotic trees and other disturbances (Giorgis et al., 2017). *Polylepis australis* sub-Andean forest and shrubland (Type 1.4) are almost restricted to deep and protected ravines and valleys above

1,800 m a.s.l., covering only 53 km² in the study area (see also Cingolani, Renison, Zak, & Cabido, 2004; Cingolani et al., 2008; Renison et al., 2015). Kurtz (1904) reported a similar distribution pattern. However, recent findings raise concern on the negative effect of fire and grazing limiting the range of this forest to such protected habitats (Renison, Hensen, Suarez, & Cingolani, 2006; Renison et al., 2015). Considering its unique flora, the relative high number of endemics and its reduced range, special attention should be paid to this vegetation type. Finally, the halophytic and sub-halophytic shrublands (Type 2.3) are restricted to azonal soils in saline depressions with severe restrictions to agriculture and therefore low probability of being threatened by conversion to agriculture.

The reduced cover of natural vegetation, supported by changing trends in socio-economic features, along with the high number of exotic species in the main vegetation types, highlight the severe degradation and fragmentation that have occurred in central Argentina, which have been particularly dramatic during the last 40 years (Carranza, Hoyos, Frate, Acosta, & Cabido, 2015; Hoyos et al., 2013). Thus, forest regeneration strategies together with the protection of the remaining stands should be carefully planned in order to increase the overall area of contiguous natural and semi-natural Chaco and Espinal forests in central Argentina. This study, reporting a synthetic description of actual vegetation types and their spatial distribution, offers a sound basis to fulfill this ambitious goal. The vegetation classification scheme proposed in this study improves current knowledge of these highly threatened forests and constitutes a first step towards the identification of efficient conservation and management strategies in central Argentina. Further studies should address extending the classification of vegetation to other Chaco and Espinal areas in Argentina and also neighbour countries, with the aim of deepening our knowledge on the distribution and dynamics of plant communities at a more detailed scale.

ACKNOWLEDGEMENTS

The authors wish to acknowledge the assistance of the following institutions: the National Scientific & Technological Research Council (CONICET), the National University of Córdoba (SECyT - UNC) and the Córdoba Province Ministry of Science and Technology (MINCyT) (Projects PICTO and PÍODO). A.T.R. Acosta was supported by the Argentinean Ministry of Science and Technology (MINCyT), fellow C. Milstein, Raíces Program. We thank the Editor and two anonymous reviewers who made valuable suggestions and commented on the first version of the manuscript. The authors declare no conflict of interest related to this publication. MC and MAG are researchers of CONICET, Argentina; SRZ is a post-doctoral fellow of CONICET; MLC is a professor at the Universidad del Molise, Italy; ATRA is a professor at the Universidad di Roma 3, Italy; JJC is a professor at the National University of Río Cuarto, Argentina, and MC, SRZ, MAG and MZ are professors at the National University of Córdoba. MC, SRZ,

MLC, MAG and ATRA designed the study and planned the data analysis; MC, SRZ, MAG and JJC performed fieldwork and contributed to data collection. MC, SRZ, MLC, ATRA and MAG wrote the article. MZ and MLC contributed to vegetation mapping and satellite data processing.

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SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

APPENDIX S1 Synoptic table showing the identified woody vegetation types along with the percentage constancy and mean cover based on 351 réleves collected in Córdoba, central Argentina

APPENDIX S2 Protocol of map construction of the native woody vegetation types of Córdoba province, central Argentina

How to cite this article: Cabido M, Zeballos SR, Zak M, et al.

Native woody vegetation in central Argentina: Classification of Chaco and Espinal forests. *Appl Veg Sci*. 2018;21:298–311.

<https://doi.org/10.1111/avsc.12369>